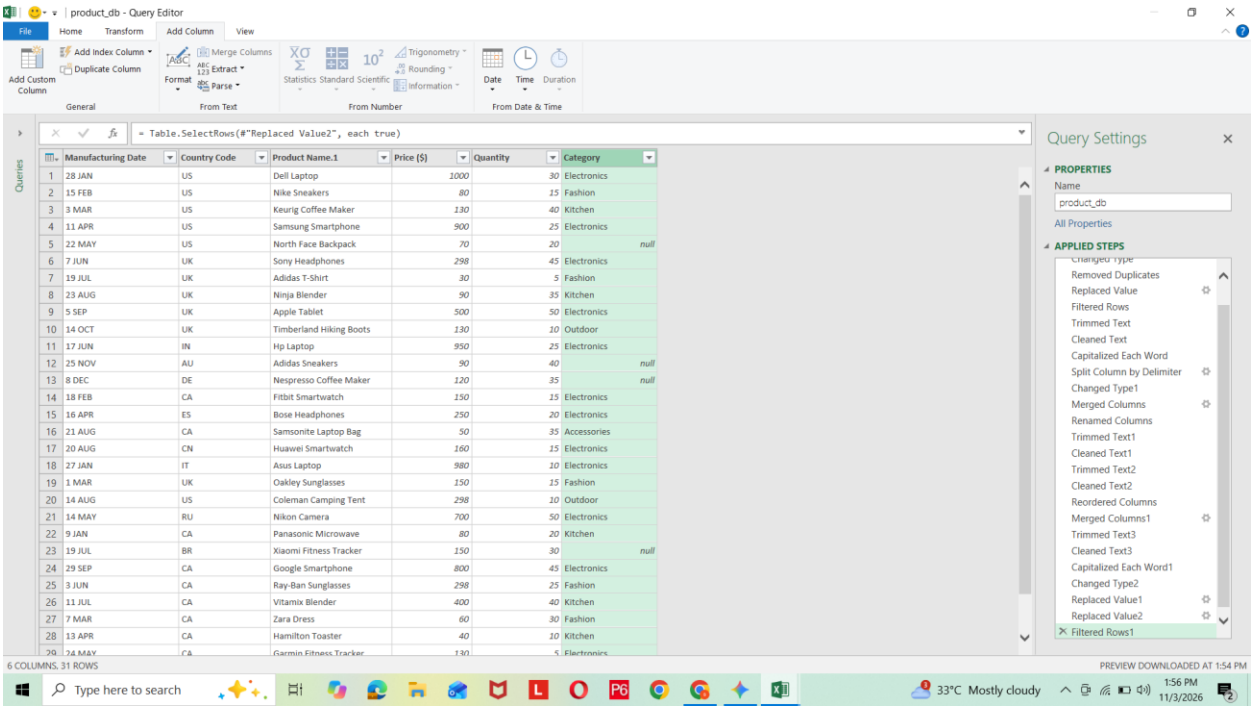
Excel Assignment 2 – Data Exploration & Cleaning Report

Prepared by: Rasika Thiruppathi

Course: AI-Driven Data Analytics

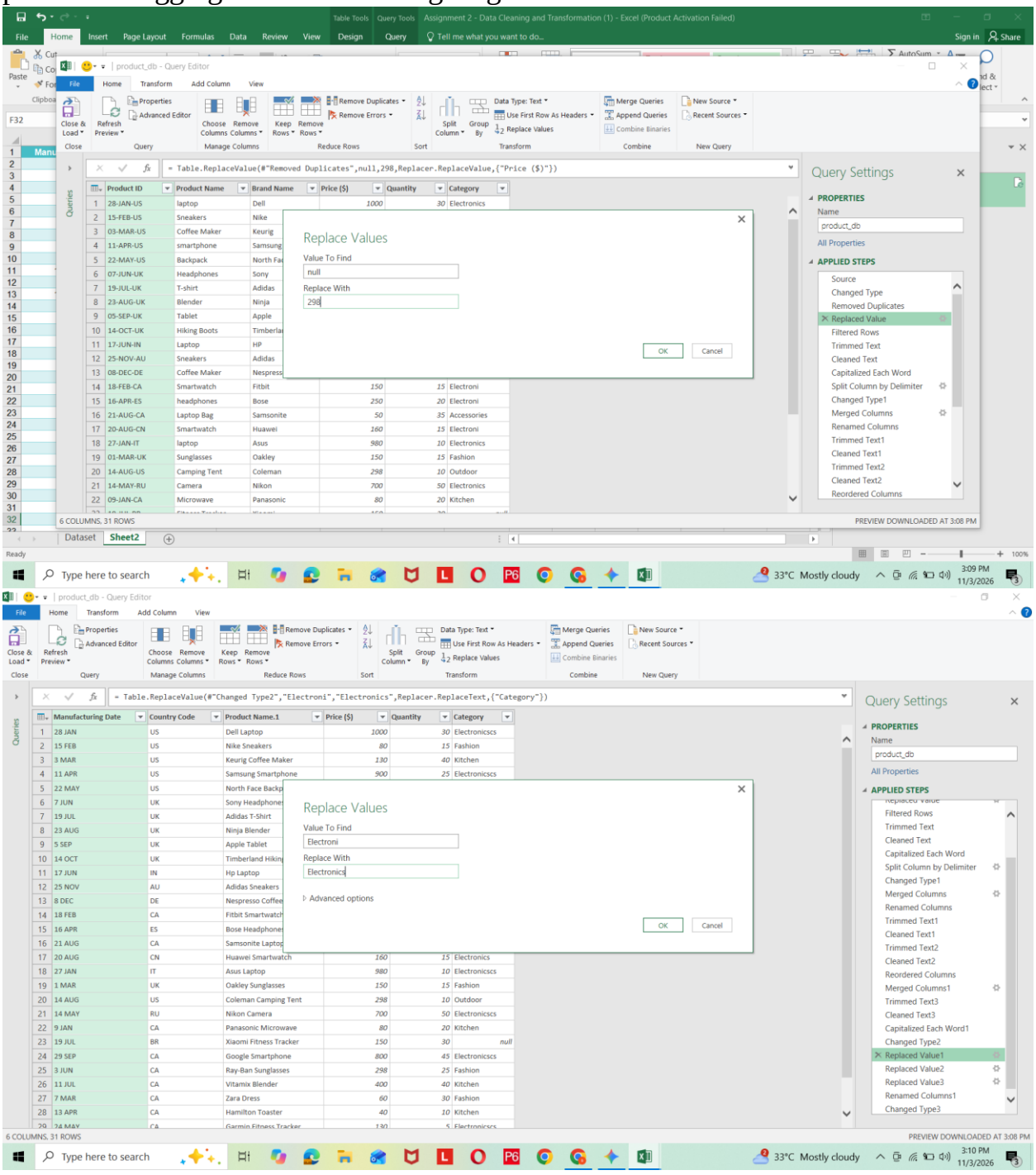
Executive Summary

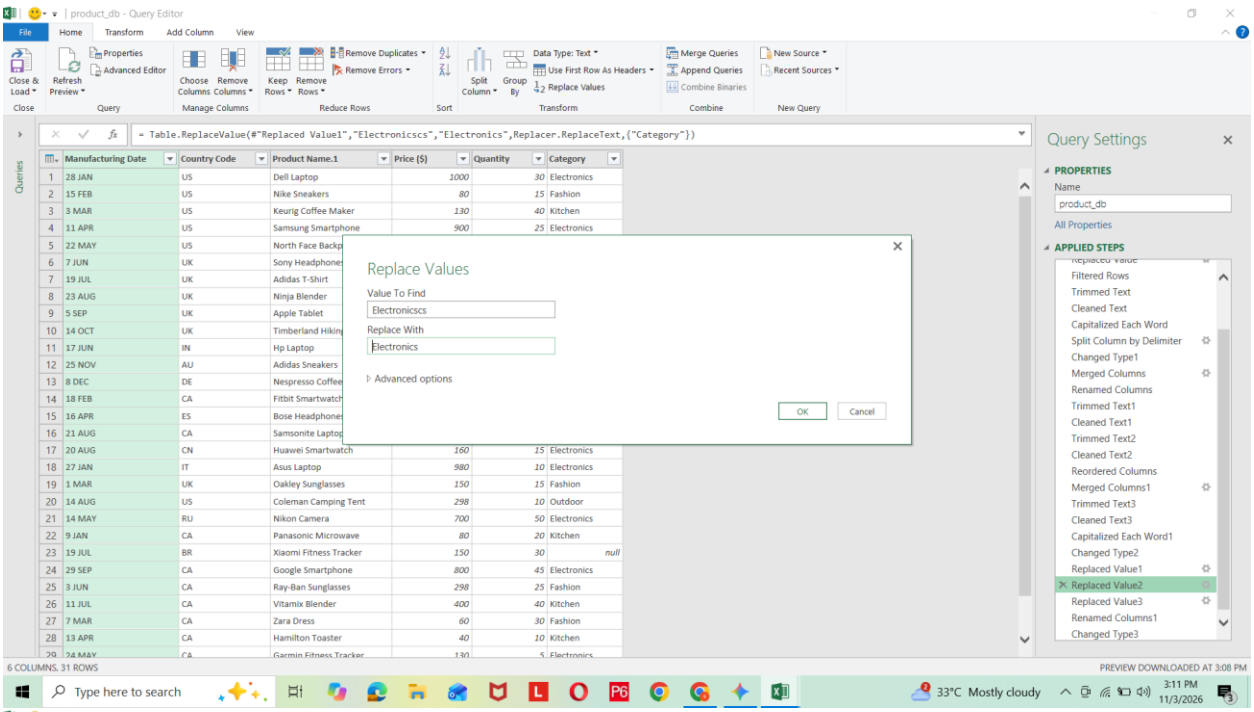
This document details the step-by-step data transformation and cleaning workflow executed on the **product_db** dataset using Power Query and Microsoft Excel. The primary goal was to resolve missing values, standardize text formatting, eliminate duplicates, extract and merge core product attributes, and apply visual formatting rules for business reporting.

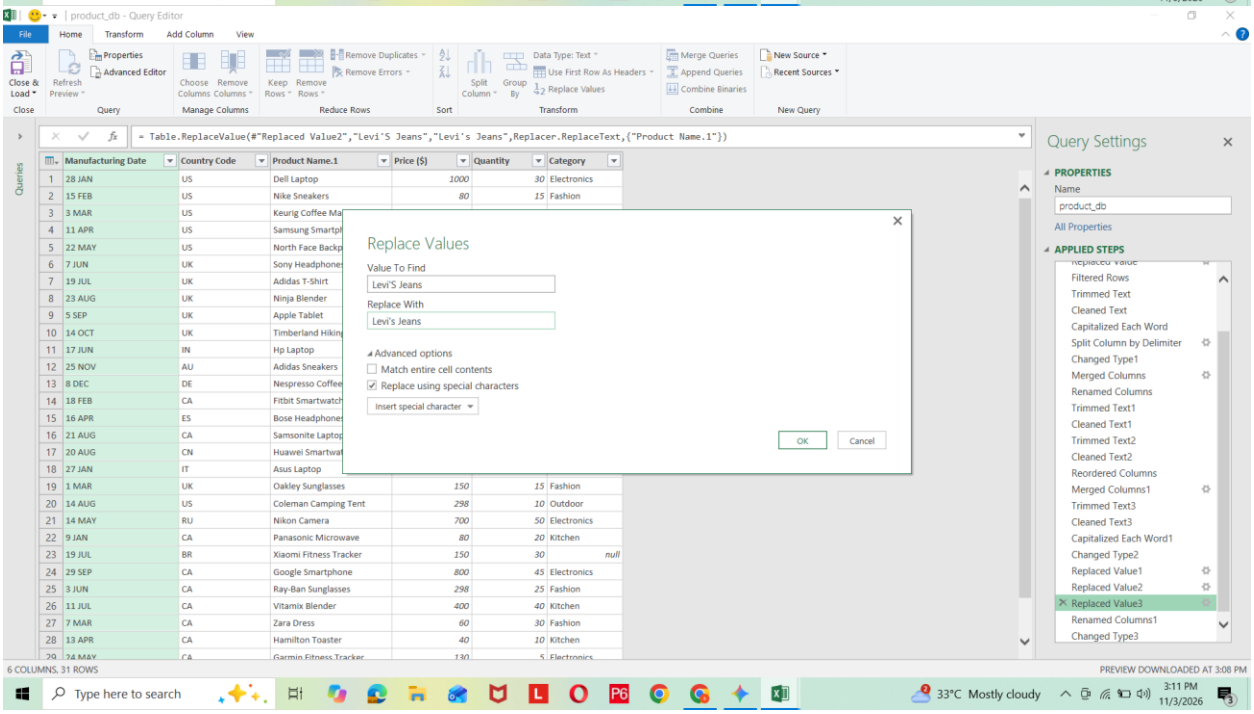


1. Find and Replace Corrections

Text formatting inconsistencies across product and brand attributes were identified and standardized using the **Replace Values** and text cleaning tools in Power Query. Casing inconsistencies and extra whitespace were cleaned to ensure full dataset uniformity, preventing misgrouping or duplication during pivot table aggregation and filtering stages.

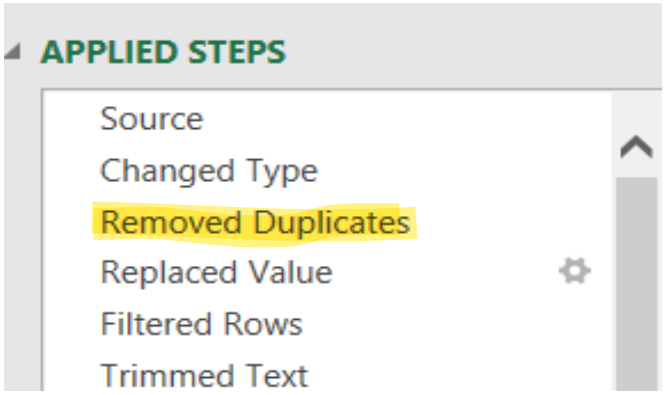



product_db - Query Editor
File
Home
Transform
Add Column
View
= Table.ReplaceValue(#"Replaced Value1","Electronicscs","Electronics",Replacer.ReplaceText,{"Category"})
Replace Values
Value To Find
Electronicscs
Replace With
Electronics
Advanced options
OK
Cancel
Query Settings
PROPERTIES
Name
product_db
All Properties
APPLIED STEPS
Filtered Rows
Trimmed Text
Cleaned Text
Capitalized Each Word
Split Column by Delimiter
Changed Type1
Merged Columns
Renamed Columns
Trimmed Text1
Cleaned Text1
Trimmed Text2
Cleaned Text2
Reordered Columns
Merged Columns1
Trimmed Text3
Cleaned Text3
Capitalized Each Word1
Changed Type2
Replaced Value1
Replaced Value2
Replaced Value3
Renamed Columns1
Changed Type3
6 COLUMNS, 31 ROWS
PREVIEW DOWNLOADED AT 3:08 PM
Type here to search
33°C Mostly cloudy
3:11 PM
11/3/2026


product_db - Query Editor
File
Home
Transform
Add Column
View
= Table.ReplaceValue(#"Replaced Value2","Levi'S Jeans","Levi's Jeans",Replacer.ReplaceText,{"Product Name.1"})
Replace Values
Value To Find
Levi'S Jeans
Replace With
Levi's Jeans
Advanced options
Match entire cell contents
Replace using special characters
Insert special character
OK
Cancel
Query Settings
PROPERTIES
Name
product_db
All Properties
APPLIED STEPS
Filtered Rows
Trimmed Text
Cleaned Text
Capitalized Each Word
Split Column by Delimiter
Changed Type1
Merged Columns
Renamed Columns
Trimmed Text1
Cleaned Text1
Trimmed Text2
Cleaned Text2
Reordered Columns
Merged Columns1
Trimmed Text3
Cleaned Text3
Capitalized Each Word1
Changed Type2
Replaced Value1
Replaced Value2
Replaced Value3
Renamed Columns1
Changed Type3
6 COLUMNS, 31 ROWS
PREVIEW DOWNLOADED AT 3:08 PM
Type here to search
33°C Mostly cloudy
3:11 PM
11/3/2026

2. Duplicate Removal Process

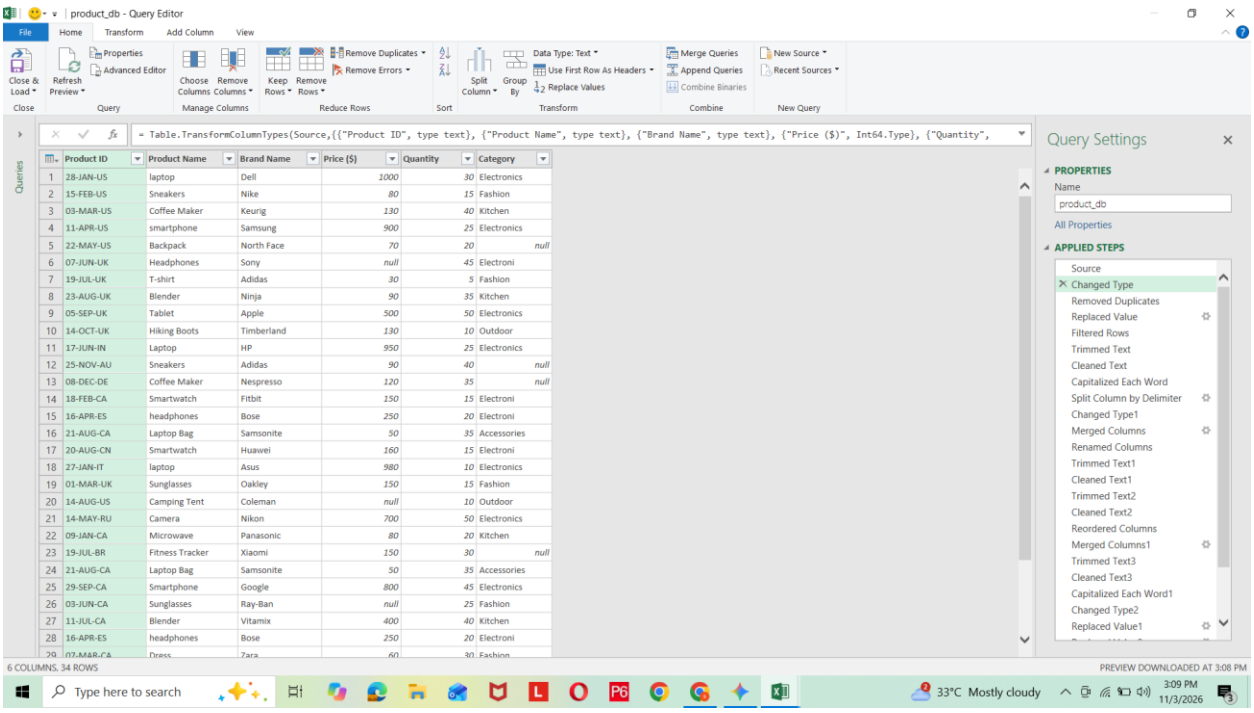
The dataset was audited for redundant rows by evaluating unique record keys. Executing the **Remove Duplicates** transformation removed all identical entries. This ensures data integrity and prevents metric inflation (such as total sales or average prices) in downstream reporting.



3. Column Splitting and Merging

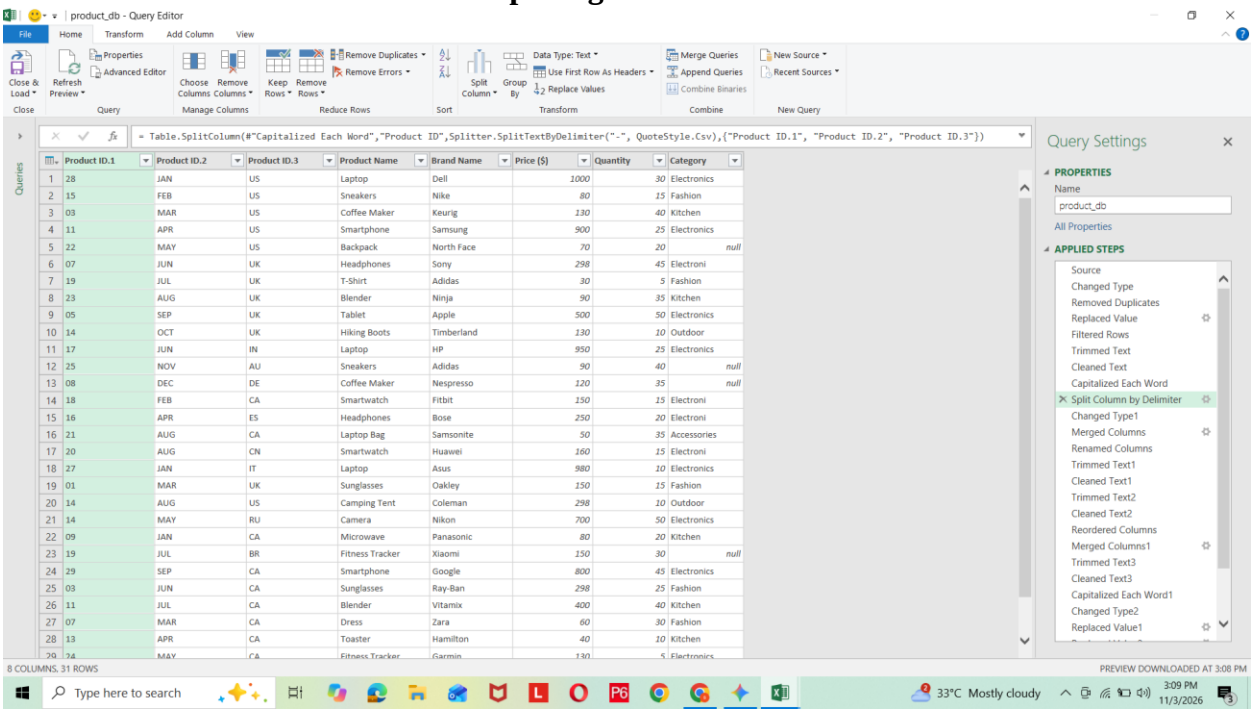
 Column Splitting: The composite **Product ID** string (e.g., 28-JAN-US) was split using hyphens as delimiters to isolate the day, month, and country code into distinct attributes.

 Column Merging: The **Brand** and **Product Name** fields were combined into a unified **Product Brand** column with space separation to create a cleaner, more readable primary text label for Excel reporting.



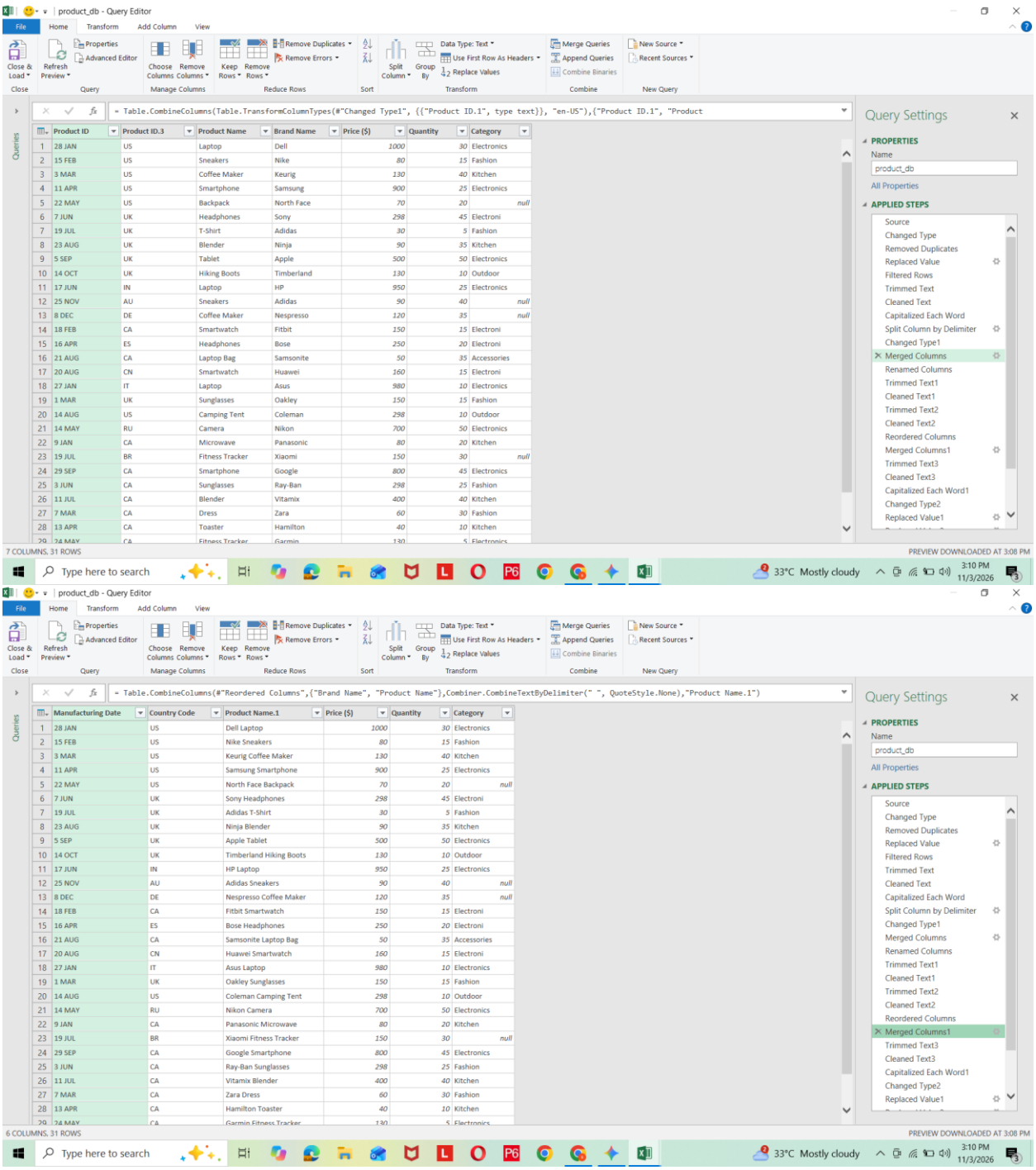
	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
1	28-JAN-US	laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	null
6	07-JUN-UK	Headphones	Sony	null	45	Electroni
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	null
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
15	16-APR-ES	headphones	Bose	250	20	Electroni
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
18	27-JAN-IT	laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	null
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen
28	16-APR-ES	headphones	Bose	250	20	Electroni
29	07-MAR-CA	Dress	Zara	60	30	Fashion

Before splitting the column



	Product ID.1	Product ID.2	Product ID.3	Product Name	Brand Name	Price ($)	Quantity	Category
1	28	JAN	US	Laptop	Dell	1000	30	Electronics
2	15	FEB	US	Sneakers	Nike	80	15	Fashion
3	03	MAR	US	Coffee Maker	Keurig	130	40	Kitchen
4	11	APR	US	Smartphone	Samsung	900	25	Electronics
5	22	MAY	US	Backpack	North Face	70	20	null
6	07	JUN	UK	Headphones	Sony	298	45	Electroni
7	19	JUL	UK	T-Shirt	Adidas	30	5	Fashion
8	23	AUG	UK	Blender	Ninja	90	35	Kitchen
9	05	SEP	UK	Tablet	Apple	500	50	Electronics
10	14	OCT	UK	Hiking Boots	Timberland	130	10	Outdoor
11	17	JUN	IN	Laptop	HP	950	25	Electronics
12	25	NOV	AU	Sneakers	Adidas	90	40	null
13	08	DEC	DE	Coffee Maker	Nespresso	120	35	null
14	18	FEB	CA	Smartwatch	Fitbit	150	15	Electroni
15	16	APR	ES	Headphones	Bose	250	20	Electroni
16	21	AUG	CA	Laptop Bag	Samsonite	50	35	Accessories
17	20	AUG	CN	Smartwatch	Huawei	160	15	Electroni
18	27	JAN	IT	Laptop	Asus	980	10	Electronics
19	01	MAR	UK	Sunglasses	Oakley	150	15	Fashion
20	14	AUG	US	Camping Tent	Coleman	298	10	Outdoor
21	14	MAY	RU	Camera	Nikon	700	50	Electronics
22	09	JAN	CA	Microwave	Panasonic	80	20	Kitchen
23	19	JUL	BR	Fitness Tracker	Xiaomi	150	30	null
24	29	SEP	CA	Smartphone	Google	800	45	Electronics
25	03	JUN	CA	Sunglasses	Ray-Ban	298	25	Fashion
26	11	JUL	CA	Blender	Vitamix	400	40	Kitchen
27	07	MAR	CA	Dress	Zara	60	30	Fashion
28	13	APR	CA	Toaster	Hamilton	40	10	Kitchen
29	24	MAY	CA	Fitness Tracker	Garmin	130	5	Electronics

After splitting up columns

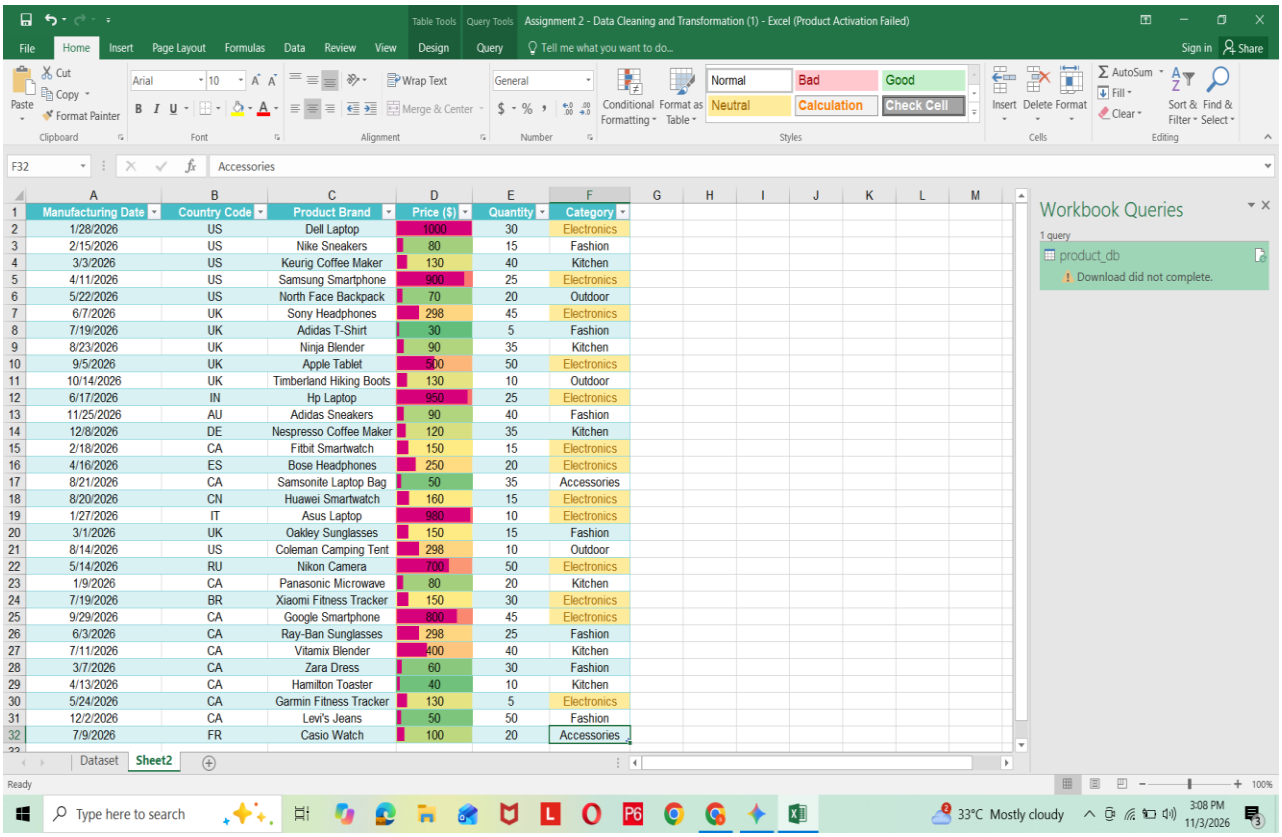


Merged columns

4. Conditional Formatting Results

Once loaded into Excel, visual formatting was applied to enhance readability:

- **Price Column:** Applied **Data Bars** (or Color Scales) to provide an immediate visual gauge of relative item costs across the dataset.
- **Category Column:** Created a custom highlight rule (Text that Contains: "Electronics") to immediately draw focus to top-priority technology inventory.

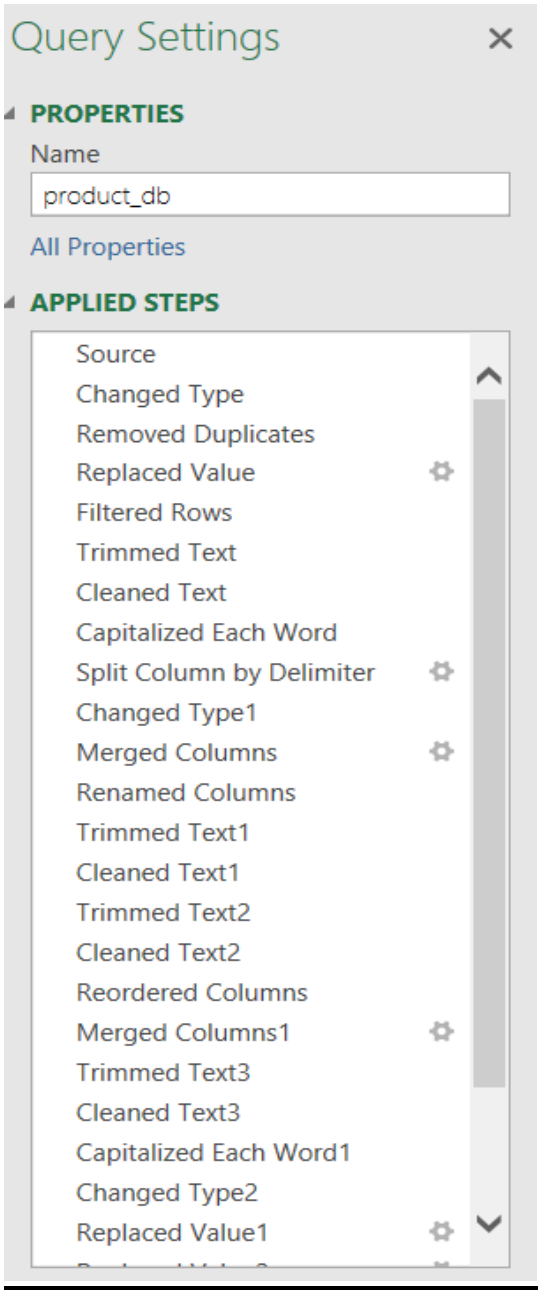


Manufacturing Date	Country Code	Product Brand	Price ($)	Quantity	Category
1/28/2026	US	Dell Laptop	1000	30	Electronics
2/15/2026	US	Nike Sneakers	80	15	Fashion
3/3/2026	US	Keurig Coffee Maker	130	40	Kitchen
4/11/2026	US	Samsung Smartphone	900	25	Electronics
5/22/2026	US	North Face Backpack	70	20	Outdoor
6/7/2026	UK	Sony Headphones	298	45	Electronics
7/19/2026	UK	Adidas T-Shirt	30	5	Fashion
8/23/2026	UK	Ninja Blender	90	35	Kitchen
9/5/2026	UK	Apple Tablet	500	50	Electronics
10/14/2026	UK	Timberland Hiking Boots	130	10	Outdoor
6/17/2026	IN	Hp Laptop	950	25	Electronics
11/25/2026	AU	Adidas Sneakers	90	40	Fashion
12/8/2026	DE	Nespresso Coffee Maker	120	35	Kitchen
2/18/2026	CA	Fitbit Smartwatch	150	15	Electronics
4/16/2026	ES	Bose Headphones	250	20	Electronics
8/21/2026	CA	Samsonite Laptop Bag	50	35	Accessories
8/20/2026	CN	Huawei Smartwatch	160	15	Electronics
1/27/2026	IT	Asus Laptop	980	10	Electronics
3/1/2026	UK	Oakley Sunglasses	150	15	Fashion
8/14/2026	US	Coleman Camping Tent	298	10	Outdoor
5/14/2026	RU	Nikon Camera	700	50	Electronics
1/9/2026	CA	Panasonic Microwave	80	20	Kitchen
7/19/2026	BR	Xiaomi Fitness Tracker	150	30	Electronics
9/29/2026	CA	Google Smartphone	800	45	Electronics
6/3/2026	CA	Ray-Ban Sunglasses	298	25	Fashion
7/11/2026	CA	Vitamix Blender	400	40	Kitchen
3/7/2026	CA	Zara Dress	60	30	Fashion
4/13/2026	CA	Hamilton Toaster	40	10	Kitchen
5/24/2026	CA	Garmin Fitness Tracker	130	5	Electronics
12/2/2026	CA	Levi's Jeans	50	50	Fashion
7/9/2026	FR	Casio Watch	100	20	Accessories

5. Missing Value Handling

Missing (null) entries in the **Category** column were identified during data exploration and manually reviewed. Each blank cell was manually populated with its correct classification (such as *Electronics*, *Fashion*, or *Kitchen*) based on the corresponding product description. This hands-on cleaning step ensured complete data integrity and proper categorization for every item prior to final reporting.

6. Steps applied in power query



FINAL CLEANED AND FORMATTED REPORT

Manufacturing Date	Country Code	Product Brand	Price ($)	Quantity	Category
1/28/2026	US	Dell Laptop	1000	30	Electronics
2/15/2026	US	Nike Sneakers	80	15	Fashion
3/3/2026	US	Keurig Coffee Maker	130	40	Kitchen
4/11/2026	US	Samsung Smartphone	900	25	Electronics
5/22/2026	US	North Face Backpack	70	20	
6/7/2026	UK	Sony Headphones	298	45	Electronics
7/19/2026	UK	Adidas T-Shirt	30	5	Fashion
8/23/2026	UK	Ninja Blender	90	35	Kitchen
9/5/2026	UK	Apple Tablet	500	50	Electronics
10/14/2026	UK	Timberland Hiking Boots	130	10	Outdoor
6/17/2026	IN	Hp Laptop	950	25	Electronics
11/25/2026	AU	Adidas Sneakers	90	40	
12/8/2026	DE	Nespresso Coffee Maker	120	35	
2/18/2026	CA	Fitbit Smartwatch	150	15	Electronics
4/16/2026	ES	Bose Headphones	250	20	Electronics
8/21/2026	CA	Samsonite Laptop Bag	50	35	Accessories
8/20/2026	CN	Huawei Smartwatch	160	15	Electronics
1/27/2026	IT	Asus Laptop	980	10	Electronics
3/1/2026	UK	Oakley Sunglasses	150	15	Fashion
8/14/2026	US	Coleman Camping Tent	298	10	Outdoor
5/14/2026	RU	Nikon Camera	700	50	Electronics
1/9/2026	CA	Panasonic Microwave	80	20	Kitchen
7/19/2026	BR	Xiaomi Fitness Tracker	150	30	
9/29/2026	CA	Google Smartphone	800	45	Electronics
6/3/2026	CA	Ray-Ban Sunglasses	298	25	Fashion
7/11/2026	CA	Vitamix Blender	400	40	Kitchen
3/7/2026	CA	Zara Dress	60	30	Fashion
4/13/2026	CA	Hamilton Toaster	40	10	Kitchen
5/24/2026	CA	Garmin Fitness Tracker	130	5	Electronics
12/2/2026	CA	Levi's Jeans	50	50	Fashion
7/9/2026	FR	Casio Watch	100	20	Accessories